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Methodology for enabling Digital Twin using advanced physics-based modelling in predictive maintenance

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Abstract

This paper presents a methodology for advanced physics-based modeling aiming to enable the Digital Twin (DT) concept in predictive maintenance applications. The proposed methodology consists of two main pillars; Digital model creation and Digital Twin enabling. Following this procedure, which is analyzed in the paper, the user is able to define, create and utilize the digital model of a resource, as well as to enable the DT concept. A digital model of an industrial robot was created aiming to validate the proposed methodology which can be utilized potentially in predictive maintenance applications.

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1. Introduction

Modern industries rely on machine tools of high complexity, which makes the maintenance scheduling a troubling task. Maintenance scheduling would be easier if the health status of each piece of equipment were known at every time instant. Through the Digital Twin (DT) concept, one could have an accurate estimation of the equipment's condition. This is the reason why the Digital Twin concept appears more and more in industrial applications related to predictive maintenance.

The research leading to DT concept started from Model-based Maintenance, which has been discussed for years. It has been argued that faults and deterioration of mechanisms can be described using Function-Behavior-State modelling [1]. Prognostic models have been constructed based on different random load conditions, while the hidden damage has been estimated using interacting multiple models and the remaining life estimation has been performed by mixing model-based life predictions via time averaged mode probabilities [2]. The modelling of fatigue-damage and the structural health

monitoring have been combined to develop a computational framework for fatigue-damage prediction in foil-scale laminated composite structures [3]. Virtual Commissioning's effects on verifying the mechanical behavior of a system and improving the real commissioning time have been investigated [4] [5]. Oppelt et al. investigated automatically generating simulation models based on plant engineering data for Virtual Commissioning (VC) [6].

One of the first definition of the Digital Twin was forged by NASA [7]. In terms of manufacturing, Negri et al. defined DT as follows: "*The DT consists of a virtual representation of a production system that is able to run on different simulation disciplines that is characterized by the synchronization between the virtual and a real system, thanks to sensed data and connected smart devices, mathematical models and real time data elaboration.*". In the context of Industry 4.0, DT can be used for forecasting and optimization of production systems at each life cycle phase in real time [8]. The DT concept model contains three parts; the physical object in real space, the virtual object in virtual space, and the flows of data and information that ties the virtual and real products together

[9]. The data flows between the existing physical object and the digital object are fully integrated in both directions [10].

Some researchers believe that a Digital Twin should be an ultra-high fidelity simulation [20]. This approach leads to models which are usually hard to implement in manufacturing, because they demand high computational effort, expertise from the part of the model creator as well as the user's, and they are time consuming to build. Others argue that the Digital Twin should be a reasonable estimation of the product operation [21], risking the model accuracy.

Extensive research has been conducted on Digital Twin for aircrafts' applications, e.g. to assess an aircraft's ability to meet mission requirements, using a CFD model and a FEM [11] and others[12][13]. AutomationML has been proposed for modelling attributes related to DT [19]. M. Grieves suggested an architecture for the DT consisting of the physical entity, the virtual model and their interaction aiming to compare the manufactured to the designed product [9]. Based on this architecture an extended five dimensional has been proposed from Tao et al. for prognostics and health management [14]. It has also been tried to extend 3D-CAD model through behavioral information to create a Digital Twin [15]. An approach for predicting the condition and the status of machines as a result of simulations without ceasing the operation of the machines has also been presented [16]. a Digital Twin for corrective maintenance has been developed in OpenModelica, using an observable fault block for distance sensor value and corresponding error classification in the simulation [18]. Another research, in the same path but exploiting cloud databases has been conducted, as well [17].

Although extended research has been conducted on maintenance and remaining life prediction by using the Digital Twin method, there is still no easily adaptable methodology to enable the creation of physics-based models for all kinds of equipment through a fully defined procedure. As physics-based model, it refers to a simplified description, especially a mathematical one, of a system or process, to assist calculations and predictions consisted of an hierarchical structure of components and sub-components representing physical phenomena and connection lines among them to represent the actual physical coupling. There is no common line for the development and implementation of the Digital Twin concept, necessary to enable adoption of Digital Twin technology for maintenance activities in production plants. Because each machine will have its own digital model and there is great variety of machines in modern production plants, the need for a common framework for the creation of the Digital Twin in different industrial sectors is imperative.

The proposed approach is as generic as possible to ensure the modelling of different types of resources and grant Digital Twins, balancing computational effort and accuracy. It is also based on three phases to reduce modelling effort and deal with the high complexity and amount of machine components without resulting in extremely detailed models. The first phase concerns the modelling of the kinematic and dynamic behavior of a machine, the second is for modelling virtual sensors to gather data during the simulation, and the third concerns the identification of a set of parameters to be updated

for ensuring the digital model represents the real condition of the modelled resource.

This paper focuses on presenting the methodology for enabling Digital Twin using advanced physics-based modelling. In section 2 the approach is described in detail, in section 3 the implementation of this approach is presented based on a real industrial pilot case. The results of the implementation are presented in section 5, while the subject of the section 6 is the conclusions and the future work.

2. Approach

The creation of digital models can be a demanding task due to the intricate design of modern machines. Due to the high importance of the machines' availability in the production plants, it is required to define a generic modelling methodology [22]. In order to reduce the modelling efforts and provide a common framework for the modeling of different resources enabling Digital Twins, three phases of modelling are suggested (Fig. 1). Each phase can be divided into actions and sub-actions.

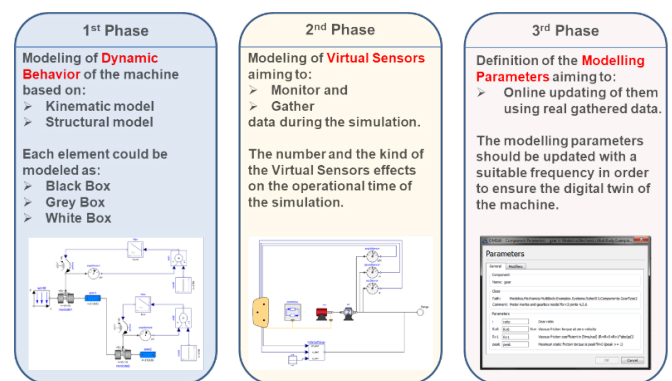


Fig. 1. Three phases of creating the advanced physics-based model.

2.1. Phase 1: Machine modelling

In this phase the selection of the machine components to be modelled, as well as their corresponding level of modelling are made. Each digital component model should cover the main physical phenomena which considerably affect the machine's behavior (static deformations, vibrations or backlash). These phenomena will be influenced by the degradation of the health status of the modelled resource and as a result they can be used in context of predictive maintenance. As for the modelling level, black box (without any knowledge of the internal operation), grey box (theoretical data are used to complete its model) or white box (fully described component) approaches are employed. The last step of the first phase is the composition of the components, in order to simulate the dynamic behavior of the machine. The actions of phase 1 are:

1. The definition of the machine's components that have to be modelled. This task involves the identification of machine's components which are crucial for the system's functionality and are subjected to continuous wear. The

model of non-crucial sub-systems is avoided to keep the complexity of the digital model low.

2. The definition of the modelling level of each machine's component. The modelling level is determined by the available data for the parametrization of the model. Ideally, all machines would be modelled as white boxes, but due to lack of data some components are modelled as grey boxes and some others as black. Another reason that leads to the usage of grey and black boxes is that white box modelling requires more time and increases the number of model parameters.
3. The creation of the machine model and the composition of all component models to build the complete digital machine model. This is the stage where the kinematic chain of the machine is taken under consideration within the connection of the individual component models. The kinematic chain of the digital model consists of rules and restrictions which aim to match the model to the actual real-world machine.

The actions and sub-actions of phase 1 are shown in Fig. 2.

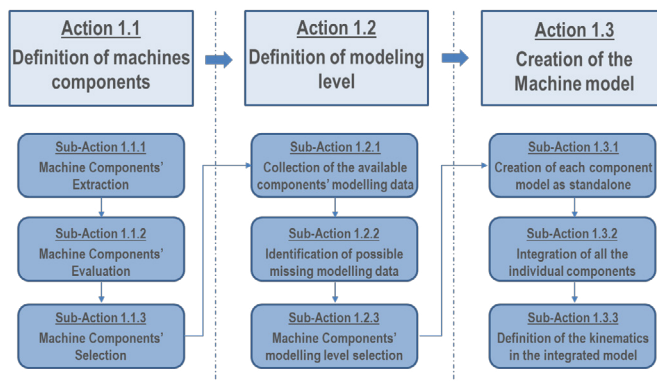


Fig. 2. Actions and sub-actions of phase 1.

2.2. Phase 2: Modelling of virtual sensors

During this phase the selection and modelling of virtual sensors takes place. The actions and sub-actions of the procedure followed is shown in Fig. 3. Virtual sensors can be modelled as a layout of elements and their functionality is to monitor and gather data from the machine's model during the simulation. In the context of predictive maintenance, it is necessary to gather data during the simulation for predicting the Remaining Useful Life (RUL). As the number of the virtual sensors is increased, the computational time of the model's simulation is also increased; so, it is important to clearly define and specify which data need to be gathered by the simulation. The actions of the second phase are:

1. The definition of data to be monitored using the virtual sensors, which involves the identification of the required data which will be used in the future and their categorization in "directly available by real sensors" and "estimated during simulation via virtual sensors", as well as the definition of the missing required data.
2. The selection of virtual sensors and the creation of their models. Firstly, the type of data monitored by the virtual sensors is identified and then the monitored components

should be selected. Finally, the creation of the components for the virtual sensors will be created.

3. The integration of the virtual sensors inside the physics-based model of the machine. The monitored data and the monitored components have been defined in previous actions, so the connection of the virtual sensors terminals inside the model is the remaining step. After that, the user will have the ability to estimate and gather data which arise during the model's simulation.

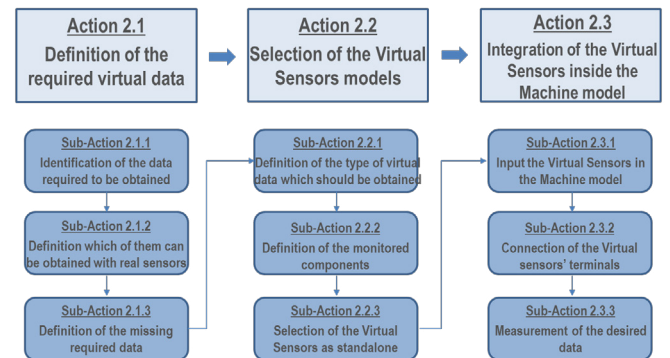


Fig. 3. Actions and sub-actions of phase 2.

2.3. Phase 3: Definition of updatable modelling parameters

The main task of phase 3 is to define the modelling parameters which will be updated in order to adjust the model to the real machine's behavior based on the controller and sensor data. These parameters will be editable and will be associated with synchronous simulation tuning. The actions of phase 3 are the following:

1. The selection of the Digital Twin components to be tuned by the synchronous simulation tuning mechanism, in order to enable the Digital Twin concept. This selection is made taking under consideration the available data coming from external sensors or machine's controller, the criticality of each component (its effect on the whole functionality of the machine), and the modelling level of each component (white, grey, black).
2. The definition of the available data to be used for tuning is a very important parameter that should be considered for the definition of the digital model to be tuned along with the source of these data.
3. The selection of the updatable modelling parameters is the final step, after the definition of the digital models which will be tuned, and the required data for the tuning. The models tuning means the tuning of a set of modelling parameters which enable the adjustment of the digital models.

The aforementioned actions and sub-actions are the main activities to create the digital model of a machine. As next step, a methodology should be utilized to enable the online tuning of the digital model and accordingly, to enable the Digital Twin concept. This methodology is based on the data exchanging via the real machine and the digital model of it, and the tuning of the second one according to real machine's behavior.

On the side of the real machine, machine's controller and external sensors are used to gather the required data. On the side of the machine's digital model, the virtual sensors are

used to gather the required data.

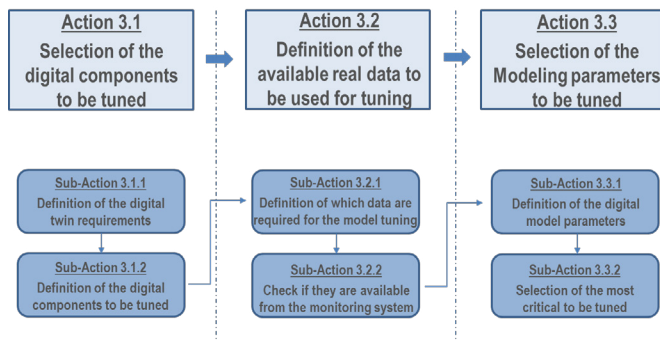


Fig. 4. Actions and sub-actions of phase 3.

On the side of the real machine, machine's controller and external sensors are used to gather the required data. On the side of the machine's digital model, the virtual sensors are used to gather the required data.

The gathered data from these two sources are structured, analyzed and compared aiming to update the updatable modelling parameters which are defined in Phase 3 of the proposed methodology. The data structuring and analysis differs among different type of machines. To implement the proposed methodology, an initialization of the modelling parameters is required. For this occasion, the manufacturer data is used.

3. Implementation

In order to successfully implement the proposed approach, it is necessary to model the critical components of the resource under study, to model the required virtual sensors to monitor data during the simulation, and to be able to define editable parameters to be able to enable the DT. Besides the modelling, it is also required to be able to structure and analyze data, as well as to process signals coming from the sensors.

The implementation of the approach employs two software packages. The first one is OpenModelica, which is a free and open-source Modelica-based modelling and simulation environment intended for modeling, simulating, optimizing and analyzing complex dynamic systems. The second one is MATLAB, which is a multi-paradigm numerical computing environment.

OpenModelica is the environment where the model of the machine is created. To create the model the OpenModelica Editor (OMEdit) is used. OMEdit is an open source graphical user interface for creating, editing and simulating Modelica models either in graphical or in textual mode. The strong advantage of OMEdit is its provision of libraries with ready components and blocks for electrical, magnetic, mechanics, fluid and thermal systems which can be used to create a complete model. Thus, with low modelling effort there is the ability to model various machines. The OMEdit communicates with the OpenModelica Compiler (OMC) which creates model/connection diagrams based on the Modelica annotations.

On the other hand, MATLAB is used for signal processing, data structuring and analysis. This software is chosen because it provides libraries with ready filtering and smoothing algorithms. Moreover, its programming language is suitable for engineering applications.

In order to enable the Digital Twin concept, the simulation's outcome should be comparable to the data coming from the real machine aiming to fine tune the digital model in real time. Thus, after the model creation and model simulation in OpenModelica, MATLAB is used to filter, structure and analyze the simulation results. Data coming from the robot sensors are also filtered, structured and analyzed.

4. Case study

The methodology was applied for the creation of an advanced physics-based model of an industrial robot. The phases 1 to 3 were followed and then the data structuring took place.

More precisely, the robot's sub-systems and components were defined with the assistance of the CAD model of the robot. Each sub-system was analyzed to its components, and each component was analyzed to its elements. Then each component's influence on the health status of the robot was evaluated. According to this evaluation, the sub-systems and components that were about to be modelled were selected. Historical data of the robot's maintenance tasks assisted the evaluation procedure. Except for the selected components, some more were modelled to serve the operation of the selected ones. After the selection of the modelled sub-systems and components the definition of their modelling level was made according to the available components' modelling data (structural, kinematic, dynamic). The components of the model are: the mechanical structure, the gearboxes, the actuators and the input signals components.



Fig. 5. Industrial robot to be modelled.

The main elements used to represent the mechanical structure of the robot were the joint (represents rotation of a rigid body around one axis) and the link (represents rigid bodies) components. The model consists of 7 links connected to each other through joints. Their combination along with the mass properties forms the kinematic model of the robot. The robot consists of 6 axis. These are modeled in the digital model as 6 revolute joints. Parameters such as length and mass of the links were defined in this phase, same as the

orientation of the base frame of the robot. Since all the required modelling data are provided either by the manufacturer, the CAD files or real time measurements the mechanical structure was modelled as a “white box”.

Gearbox consists of an ideal gear (represents the gear ratio of each axis), a bearing (represents Coulomb friction) and an inertia (represents the gears) component. Although it is the most important component of the model, because wear effects are mostly observed in this part, not all the required modelling data are available, so it was modelled as a “grey box”.

The actuator translates the input angle signal to an output torque signal and it was used as a substitute of the motor component to simplify the model. The actuator was modelled as “black box”.

The input signals component consists of a data set of robot’s joint angles and it was used to input the desired trajectory for the simulation. The input signals component is a “black box” model.

As for the combination of the components, the input signals components are connected to the actuators of each axis. The actuators create a torque signal for the motion of the corresponding link, according to the input signal. The torque signals are the inputs of the gearboxes, which are connected to the actuators and then each gearbox is connected to the corresponding link.

After the actions of phase 1, virtual sensors were modelled and integrated in the digital model for gathering data which cannot be collected by some sensorial system or the robot’s controller. The added virtual sensors collect: position data at the output shaft of the gearbox relative to the position signal before the bearing, angular speed data at the output shaft of the gearbox relative to the velocity signal before the bearing, and acceleration data at the output shaft of the gearbox relative to the acceleration signal before the affection of axis inertia. The three sensors were integrated in each gearbox of the digital model.

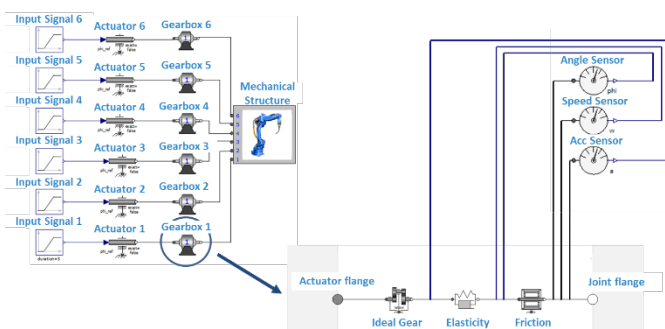


Fig. 6. Digital model of industrial robot.

As a final step according to the criteria mentioned in the approach section, two parameters were selected for the tuning of the model. The modelling parameter F_c which corresponds to the friction component of a robot gearbox and it is relative with the Coulomb friction and the parameter J_m which corresponds to gear’s inertia was selected to be tuned.

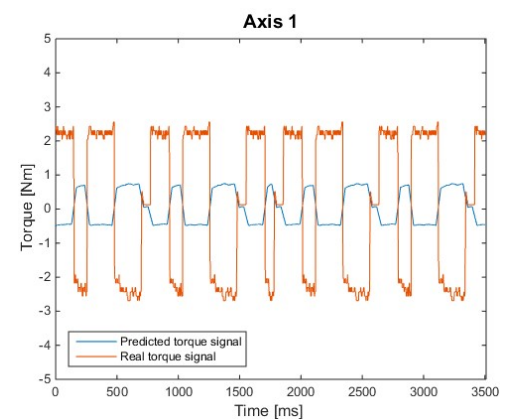
The data coming from the virtual sensors of the digital model and the data coming from real robot controller are gathered and structured aiming to identify potential deviation

between the real and the digital robot and to compensate them aiming to enable the DT concept. This compensation was achieved by the updating of the modelling parameters of the robot’s digital model.

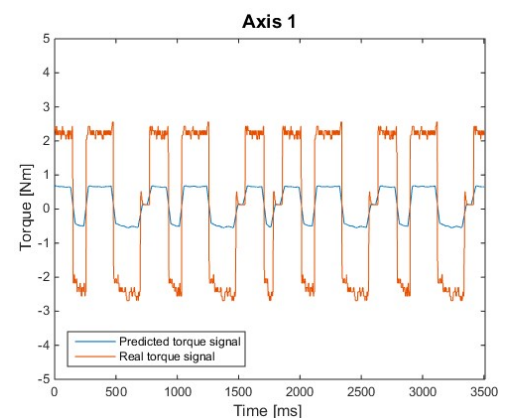
In this case study the tuning mechanism of the modelling parameter for each digital robot’s gearbox based on the comparison of the real and the predicted torque signals. The real torque signals were gathered by robot controller while the predicted one, arise by the implementation of the second-order, ordinary system of differential equation for the dynamic behaviour of a serial manipulator and applying the virtual sensors position, velocity and acceleration data.

A simulation of the model was performed, and the output signals of the simulation were compared to the ones measured from the real robot. If the deviation between the real and the predicted signals was out of the predefined threshold, the modelling parameters were tuned accordingly in order to achieve the most accurate modelling possible. The estimation of the dynamic parameters of the model is based on the aforementioned Nonlinear Least Square Method. Using this method, the value ranges for parameters were restricted significantly. An iterative process was executing using the estimated parameters with aim to evaluate behavior of the simulated model. The simulation-comparison-tuning step was repeated several times until the desired result was achieved. Indicatively some figures from the Axis 1 gearbox tuning procedure are shown in Fig. 7. In each iteration, the values of the modelling parameters were tuned until the deviation between the real and the predicted torque signal was acceptable.

a)



b)



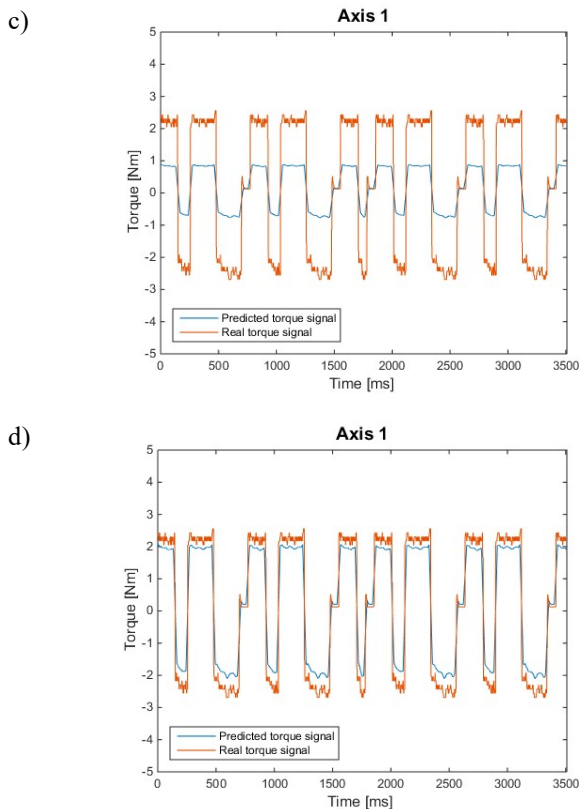


Fig. 7. Simulation-comparison-tuning step iterations 1 to 4 (a, b, c, d) until the deviation was acceptable.

5. Conclusion and Future work

The proposed approach can be utilized as a guide for creating a physics-based model to enable DT concept. This methodology can be also applied for the creation of digital models and enabling the DT concept in predictive maintenance applications. More specifically, the enabling of DT allows the utilization of the digital replicate of the real machine which in combination with degradation models can be used to calculate the RUL of each machine components for a long future period.

As future activity, the authors plan to integrate the proposed methodology in a more generic predictive maintenance framework which main role will be the evaluation of the machines' health status and the planning of the maintenance activities. Future work includes the validation of the methodology in a production plant, a replication for a different type of machine e.g. a CNC machine. Also, it is intended to improve the created models in terms of accuracy to eliminate the error between the predicted output and the real data. Finally, the tuning mechanism should be enhanced with more efficient algorithm to tune the modelling parameters. The online execution of the proposed concept with real time data exchanging will take place in the future.

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